$3d^9$

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1 $\left|3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2 3d_{xz}^\alpha\right\rangle$

Here are the CSFs:

$$|1, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\alpha\right\rangle$$
$$|2, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\alpha 3d_{x^2-y^2}^2\right\rangle$$
$$|3, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\alpha 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|4, 1/2, 1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\alpha 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|5, 1/2, 1/2\rangle = \left|3d_{xz}^\alpha 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|6, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^\beta\right\rangle$$
$$|7, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^\beta 3d_{x^2-y^2}^2\right\rangle$$
$$|8, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^2 3d_{xy}^\beta 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|9, 1/2, -1/2\rangle = \left|3d_{xz}^2 3d_{yz}^\beta 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$
$$|10, 1/2, -1/2\rangle = \left|3d_{xz}^\beta 3d_{yz}^2 3d_{xy}^2 3d_{z^2}^2 3d_{x^2-y^2}^2\right\rangle$$

The Ground State CSFs are 5 10

$\left\langle 3, \frac{1}{2}, \frac{1}{2} \left| L_x \right| 5, \frac{1}{2}, \frac{1}{2} \right\rangle = -I$

Δg_{xx}

$+2\zeta\Delta_{5-3}^{-1}$

$\left\langle 1, \frac{1}{2}, \frac{1}{2} \left| L_y \right| 5, \frac{1}{2}, \frac{1}{2} \right\rangle = I$
$\left\langle 2, \frac{1}{2}, \frac{1}{2} \left| L_y \right| 5, \frac{1}{2}, \frac{1}{2} \right\rangle = -\sqrt{3}I$

Δg_{yy}

$+2\zeta\Delta_{5-1}^{-1}$
$+6\zeta\Delta_{5-2}^{-1}$

$\left\langle 4, \frac{1}{2}, \frac{1}{2} \left| L_z \right| 5, \frac{1}{2}, \frac{1}{2} \right\rangle = I$

Δg_{zz}

$+2\zeta\Delta_{5-4}^{-1}$